MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
MODEL QUESTION PAPER - THEORY
STATISTICS
MG5DSESTA302-STATISTICAL COMPUTING USING PYTHON

Duration: 1.5 Hours **Maximum: 50 Marks**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions

Answer **any seven** questions. Each question carries **2 marks.**

1. Distinguish between a Python interpreter and a Python interface. [U][1]
2. What is the difference between append() and extend() methods in a list? Illustrate each with an example. [U][1]
3. Describe the significance of indentation in Python with an example. [An][1]
4. What is the advantage of a NumPy array over a Python list? [U][2]
5. What is the difference between a Pandas Series and a Pandas DataFrame? [U][2]
6. Write a single line of code to detect missing values in a DataFrame. [A][2]
7. Define a scatter plot. Give the Matplotlib command to create a simple scatter plot. [U][2]
8. What is lmplot? How is it different from regplot? [U][2]
9. What is Pseudo-Random Number? [U][3]
10. What is a t-test? Give the Python command to perform a one sample t-test. [U][4]

Part B

Short Essay Questions

Answer **any four** questions. Each question carries **6 marks**

11. What is an Integrated Development Environment (IDE)? Explain the major components of an IDE. [U][1]
12. Explain any five built-in methods used with Python lists. [U][1]
13. Load the built-in titanic dataset, using seaborn.load_dataset('titanic'). [A][2]
    Write Python code to:
    a) Create a DataFrame with columns age, fare, sex and name it as data.
    b) Display data types and shape.

c) Display age of first 5 female passengers.

Show expected output structure.

14. Create a long-format sales DataFrame using dummy values, containing Region, Quarter and Sales, with duplicate Region-Quarter combinations. Use pivot_table to obtain total sales, melt the result back to long format, and draw a Matplotlib bar chart of total sales by region with title and axes labels. [A][2]

15. Load the tips dataset using sns.load_dataset("tips"). Create a catplot with kind="count" to show the number of observations for each sex, using day to distinguish the groups. Write the Python code and explain the output. [A][2]

16. Explain the procedure for testing the equality of two population proportions using the Z-test. State the test statistic clearly. [U][4]

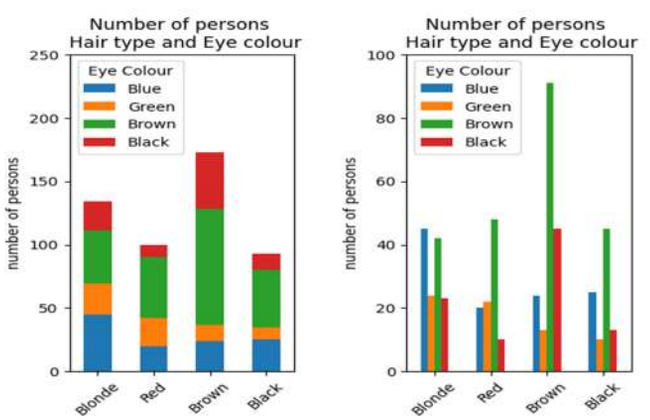
Part C

Essay questions

Answer **any one** question. Each question carries **12 marks.**

17. Write a Python programme to display the given data in sub-divided and multiple bar diagrams of the following forms. [A][2]

Hair Colours	**Eye Colours**			
	Blue	Green	Brown	Black
Blonde	45	24	42	23
Red	20	22	48	10
Brown	24	13	91	45
Black	25	10	45	13



18. Explain the complete procedure of a two-way ANOVA (with one observation per cell). Describe how the total variation is decomposed into the effects due to the two factors, their interaction, and error. Present the ANOVA table format used to summarize the results. Also give the Python codes for implementing two-way ANOVA. [U][4]